



# L4 In-Lecture Questions

Filter Banks, Multiscale Reasoning & Texture

# Q1: Frequency vs. spatial support

Two Gabor filters have the same orientation and spatial frequency. Filter B has a larger Gaussian  $\sigma$  than Filter A. Which statement is correct?

- A. Filter B has more widely spaced stripes.
- B. Filter B examines a larger neighborhood, but its stripe spacing is unchanged.
- C. The filters are identical because their frequencies are equal.
- D. Filter B detects a different orientation.

# A1: Frequency vs. spatial support (B)

- A. Frequency—not  $\sigma$ —determines the spacing between stripes.
- B. A larger  $\sigma$  widens the Gaussian envelope, so the filter examines a larger neighborhood while preserving stripe spacing.**
- C. Their envelopes and spatial support differ, so their responses need not be identical.
- D. Orientation is controlled independently of  $\sigma$ .

## Q2: Smoothing before downsampling

Why do we apply Gaussian smoothing before downsampling an image to construct a pyramid?

- A. To increase image contrast.
- B. To decrease image contrast.
- C. To suppress detail that the lower-resolution grid cannot represent reliably.
- D. To make all objects appear at the same size.

## A2: Smoothing before downsampling (C)

- A. Gaussian smoothing normally reduces contrast in fine details.
- B. Gaussian smoothing cannot increase contrast.
- C. Smoothing removes high-frequency detail and reduces aliasing during downsampling.**
- D. Objects become smaller at higher pyramid levels; they do not become equal in size.

## Q3: Pooling texture responses

Two image patches contain similar local elements, but those elements occur at slightly different positions. Why might their pooled texture vectors still be similar?

- A. Pooling summarizes which responses occur and how strongly, rather than their exact positions.
- B. Pooling preserves the exact position of every response.
- C. Pooling changes the frequencies of the filters.
- D. Pooling guarantees invariance to every rotation and scale.

## A3: Pooling texture responses (A)

- A. **The summary is less sensitive to small rearrangements of the same local patterns.**
- B. Pooling deliberately discards some positional information.
- C. Pooling summarizes responses; it does not change the filters.
- D. Rotation and scale invariance require additional design choices.